TIPA Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Virtual Manufacturing of Fibrous Materials: Structure-Properties Prediction and Interaction with Liquids			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (√), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing (√), Quantum ( )			
Industrial Technology Classification 1)	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Chemicals			
	Medium Category	Small Category	Medium Category	Small Category
	Fiber products	Non-woven fabric manufacturing		
Concept of Proposed Technology	Materials made from fibers are ubiquitous in engineering. Polymeric nonwovens are broadly used in various consumer products and as geotextiles. Fiberglass is used for insulation (thermal and acoustic) and as reinforcement for various composites. Natural fiber felt and textiles are also widely used. The fabrication of these materials faces somewhat similar challenges, some of which are addressed by the proposed work. Nonwovens are produced by spinning polymeric fibers followed by calendaring and either thermal bonding, needle punching, or spunlacing (hydroentanglement). Fiberglass is also spun, but bonding is introduced by using an adhesive mist or by soaking in an adhesive solution and subsequent vacuum absorption of the excess binder. Furthermore, the interaction of these materials with liquids has many forms: moisture wicking (hydrophobic and hydrophilic fiber mixtures), oil separation, printing/ink absorption, filtration etc. The proposed work focuses on developing computational predictive tools that relate the network			

structure to certain aspects of the material behavior, and which may be further used to control manufacturing. Figure 1 shows the types of materials and processes addressed. We will consider nonwovens and fiberglass (prioritizing one or the other function of the SME interests), which are both stochastic networks. Two types of properties will be considered in separate tasks: mechanical properties representative for the function of the respective networks, and the interaction of the network with liquids, as relevant for printing, absorption and fluid transport. The manufacturing steps to be related to properties are bonding (thermal and needlepunching of nonwovens, and adhesive bonding in fiberglass) and the interaction of the fibrous network with droplets during printing/wetting. Research on one or multiple of these processes will be prioritized as a function of the SME interests.

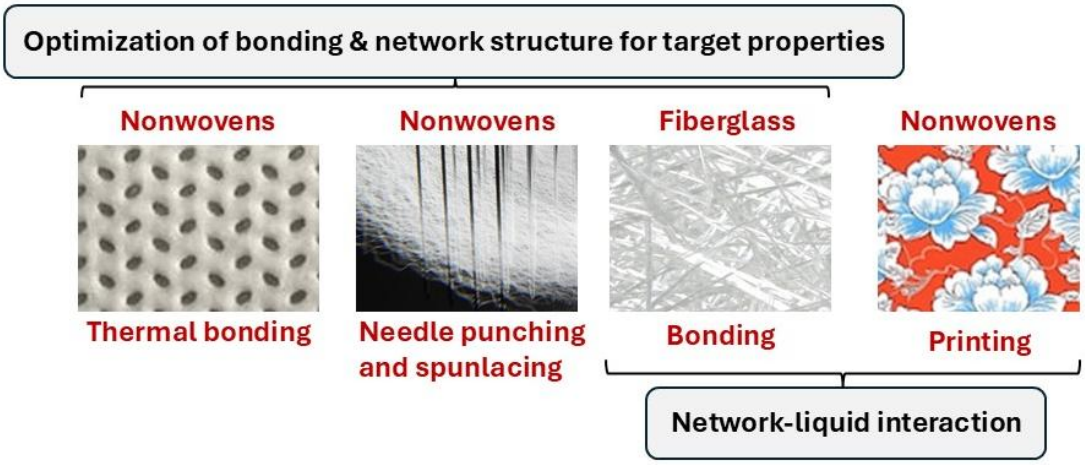

Figure 1. Types of materials, manufacturing and physical processes addressed.

Three computational tools (apps) will be developed in three tasks, as shown in Fig. 2 (orange boxes). The first tool is a predictor of mechanical properties of the network material (stiffness, strength, yield stress) based on the structure. The structure is described in terms of the fiber density (base weight), density fluctuations (flocculation), degree of crosslinking, fiber material properties, and preferential fiber orientation. The tool can be used to identify the structure that optimizes a set of material properties. To this end, an objective function is defined as a weighted sum of target properties, and the app optimizes this function in the space of defined structural features. This may be used to guide the manufacturing process indicating, for example, the desirable combination of base weight and degree of crosslinking.

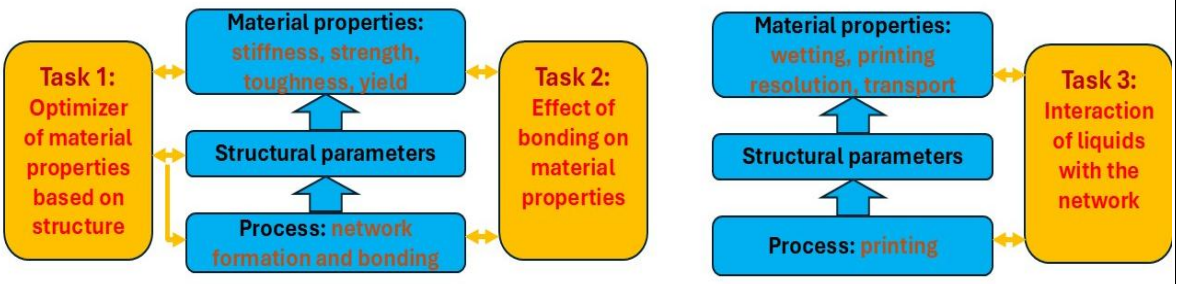

Figure 2. Structure of the proposed work.

Task 2 will develop a computational tool that relates the density of bonds (thermal, needlepunches) and the 2D pattern in which these bonds are applied to the overall material properties. This app can

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	be used to define the way the network should the bonded to achieve specific properties. Task 3 focuses on the interaction of the network with liquid droplets with specific application to printing on nonwovens and textiles, and adhesive bonding of fiberglass (to be prioritized as a function of SME interests). For given nonwoven and liquid properties (surface tension, density etc.) the app will predict the max printing resolution. Likewise, in the context of fiberglass, the app will provide an estimate of the crosslink density for given process parameters.
Rationale for Joint R&D Project	○Establish robust relations between the structure of nonwovens and/or fiberglass and the overall material properties. A need exists to improve the control of the spatial distribution of fiber density and of bonding/crosslinking density for more consistent product properties. A design tool able to predict the microstructure (structural parameters) that optimizes a user-defined set of mechanical properties is not currently available. ○Improve mechanistic understanding of network-liquid interaction. The interaction of stochastic networks of fibers with liquids is critical in many applications. The coupling of the intrinsic structural heterogeneity of the network with liquid transport, absorption and retention is insufficiently understood, particularly when the network is composed of multiple types of fibers. Better physical understanding and predictive models of these processes will be beneficial in various industrial contexts. ○Collaboration between RPI and Korean SME may evolve into a mutually beneficial synergetic partnership.
Key Words	1. Nonwovens 2. Fiberglass 3. Textiles 4. Mechanical properties 5. Fluid absorption and retention

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (5)
R&D Objectives	⬭Objective 1: Develop a computational tool that relates network structure (defined by base weight, degree of flocculation, degree of crosslinking, fiber material properties, preferential fiber orientation etc.) to mechanical properties such as stiffness, yield stress and strength. This tool can be used to optimize combinations of mechanical properties in the space of structural parameters, therefore defining a target for manufacturing. ⬭Objective 2: Develop a computational tool that relates mechanical properties of nonwovens to the type and pattern of (thermal or needlepunching) bonding while accounting for the network structure such as flocculation, preferential orientation, base weight. This tool can be used to define the bonding pattern for target properties of nonwovens. ⬭Objective 3: Develop models of the interaction of liquids with stochastic networks accounting for the network microstructure, and apply to technological processes such as printing on nonwovens and crosslinking of fiberglass.			
R&D Target Specifications	We target the development of three predictive computational tools described in section 2, while we envision prioritizing the research topics to align best with the SME interests. Specifications will depend on the application. However, we expect that (i) the tools will predict functional relations (trends, qualitative) between inputs and outputs with high accuracy, while (ii) the numerical predictions (quantitative) will be within the experimental error range (defined by the variability from sample to sample in experiments); the model and experimental levels of confidence will be comparable.			
R&D Plan	**Preliminary work:** The proposed work is based on our two decades long research on the relation between structure and mechanical properties of fibrous materials [1]. We developed expressions relating the stiffness, strength and toughness of fiber networks to the fiber and crosslink properties and network structure [2-5]. We explored issues such as damping and sound insulation in networks (e.g. fiberglass) [6], the mechanisms controlling the yield point of stochastic networks [7], the effect of preferential fiber orientation on fracture and material properties [8] etc. The proposed work aims to make this background industrially useful and relevant for applications proposed by the SME. **Proposed activities:** The proposed work encompasses 3 tasks which will be prioritized function of the SME interests, Fig. 2. **Task 1**. Develop a network structure-properties predictor and optimization tool (App A). Relating the network structure to its mechanical properties is desirable in many applications. By network structure we refer to parameters such as the nominal network density (base weight), spatial variability of the network density (flocculation amplitude and wavelength), fiber and crosslink properties, fiber preferential orientation distribution, distribution of fiber diameter and			

crosslink density. The properties of interest are the stiffness, yield point, non-linear behavior, and strength.

Scaling laws relating these properties to network parameters have been developed for some types of networks [1]. These functional relations depend, to some extent, on the network structure. To be quantitative, one must go beyond scaling laws and account for scaling constants and other factors that enter the respective functional form. These constants must be calibrated based on one or a small number of experimental tests.

The proposed computational tool (App A) will be based on these scaling laws. The functional forms will be adapted to the network type used by the SME. Furthermore, constants will be calibrated based on a small set of experiments which RPI will perform with the material provided by the SME. The tool will provide quantitative predictions of the material properties listed above.

Optimization: A specific application proposed by the SME will prioritize certain material properties. An objective function including a weighted contribution of the properties of interest will be constructed and will be optimized in the space of parameters describing the network structure. Therefore, App A will predict the parametric set optimizing the target properties. This will help adjust the manufacturing process to achieve the desired structure and hence, properties. Figure 3 shows the proposed App A function and embedding in process design.

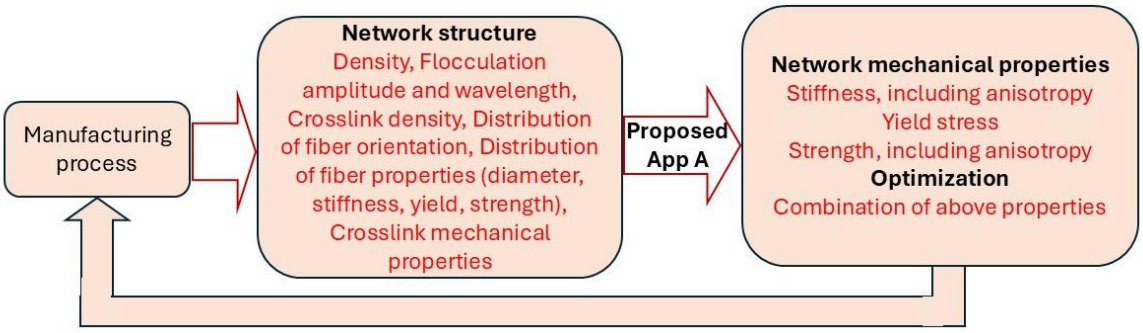

Figure 3. Proposed App A and place within the manufacturing process.

Task 2. Develop a network bonding design tool for target mechanical properties.

The creation of crosslinks in nonwovens via thermal bonding or needlepunching, and in fiberglass via adhesive spraying or soaking, is critical for the function of the material. Nonwovens are typically over-crosslinked, while the pattern in which thermal bonds are placed or needle punches are applied is not always defined based on mechanistic considerations.

We propose to develop a computational tool (App B), which predicts the network mechanical properties for given density, size and spatial distribution of thermal bonds or needle punches. This App can then be used to explore alternative ways of bonding to optimize a property or a weighted average of several material properties. Figure 4 shows the function of the proposed App B.

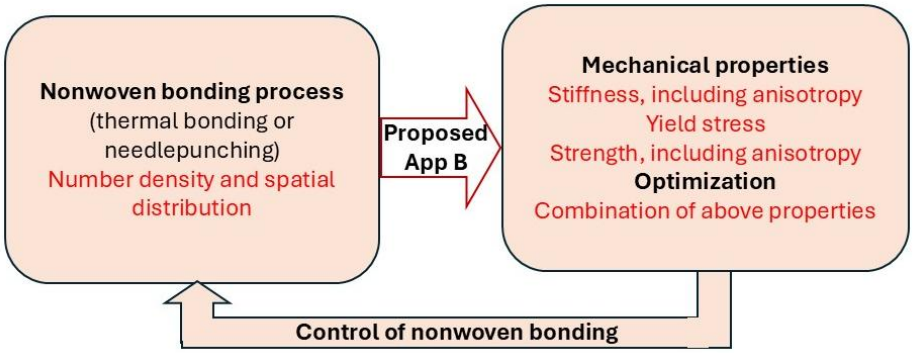

Figure 4. App B used for bonding process design.

These models will be validated based on experiments performed (at RPI or SME) with the material of interest for the SME. The specification of material properties of interest is to be done in collaboration with the SME.

Task 3. Develop a predictive tool for the interaction of networks with liquids.

In many applications networks interact with liquid droplets, examples including printing on textiles, moisture winking and absorption of fluids in various consumer products. A particularly poorly understood process is the crosslinking of fiberglass, used for acoustic and thermal insulation in automobiles, appliances and buildings, with a mist of a polymeric binder solution. The interaction of the network with the liquid is not controlled in this process and the number density of resulting crosslinks is typically unknown.

This motivates the proposed research on the interaction of liquid droplets with the networks. We will consider situations in which droplets are isolated, as this is more relevant for the applications mentioned above. The network will be considered stochastic with or without preferential fiber orientation. In separate studies, we will consider the network to be composed from hydrophilic fibers, from hydrophobic fibers and from a combination of hydrophilic and hydrophobic fibers. Coupled mechanics and transport (Lattice Boltzmann) modeling will be used to account for the interaction of the liquid with fibers. Droplets of known surface tension and size will be brought in contact with the network which, in turn, will re-organize due to the capillary action. The wetting will be characterized in terms of the spread and re-organization of the liquid volumes within the network. The process will be repeated for networks of various densities and crosslink densities, and fibers of different diameters. The number of liquid pools connecting two or multiple fibers will be determined. In the case of a composite nonwoven of fibers with different affinities for the liquid, the density and relative distribution of fibers of different properties will be considered as an additional key parameter.

The results of this analysis will lead to a data set relating the liquid distribution in the network (amount of liquid forming crosslinks, spread, total volume of liquid sustained by the network etc.) to the network structure, fiber type and composition, affinity to the liquid etc. This data set will be used to train a Machine learning (ML) model, which will become the third app proposed, App C. The app will relate system parameters that characterize the liquid and the network to the distribution of liquid within the network, therefore predicting physical quantities such as the density of crosslinks in a fiberglass mat or the spread of droplets during printing on a nonwoven. Figure 5

shows a schematic of the work to be performed in this task.

The specific application of interest for the SME will define more precisely the physics to be accounted for and the output parameters of App C.

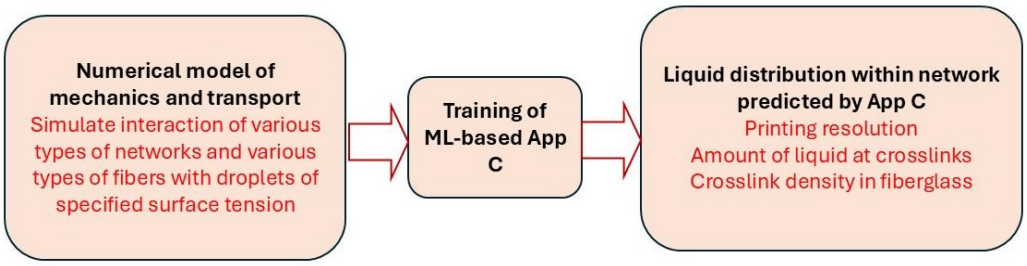

Figure 5. Schematic of the work proposed in Task 3.

Milestones and timeline:

Phase 1 (6 months): By the end of Phase 1 we will develop App A. At this stage, the app will provide qualitative relationships between material properties and structural network parameters. The app will be validated by existing numerical solutions and experimental data and will not be specifically tailored for the SME application.

Phase 2:

12 months: Task 1 will be finalized, with App A being calibrated based on Phase 1 and dedicated experimental material characterization and testing using the materials provided by the SME.

18 months: App B will be developed and calibrated to address the needs of the SME.

24 months: Validation of App B based on a small set of material characterization and testing experiments using the SME materials, hence completing Task 2. Begin investigation of Task 3.

30 months: Develop models, explore parametric space and collect data related to the interaction of liquids with the network in the context of the SME application: Task 3, part 1.

36 months: Calibrate the ML-based App C and use as process predictor for the SME application.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	⬭Rensselaer Polytechnic Institute (RPI) will develop App A. Full calibration will be performed by RPI using experimental data (obtained at RPI or SME) for the SME material. ⬭RPI will develop App B. Full calibration will be performed by RPI using experimental data (obtained at RPI or SME) for the SME material. ⬭ RPI will develop all computational models proposed in Task 3 and will calibrate the ML-based App C. RPI will discuss with the SME the best ways to validate App C, which depends on the application considered (nonwovens vs. fiberglass). ⬭ RPI and SME will work together to implement the developed apps.

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Expected R&R of Korean SME Partner	○ At the beginning of the project the SME will discuss with RPI the application(s) of interest and the associated manufacturing process. This will help prioritize the tasks. ○ The SME will provide RPI with materials for testing, which will allow full calibration and validation of Apps A and B. Alternately, SME may perform mechanical tests on samples prepared per RPI instructions. ○ The SME will interact with RPI on Task 3 to ensure research activities are closely related to the application. Validation will be performed in collaboration based on outcomes of the manufacturing process used by the SME.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	The outcomes of the proposed research are: - App A: a computational tool that relates network structure (defined by base weight, degree of flocculation, degree of crosslinking, fiber material properties, preferential fiber orientation etc.) to mechanical properties such as stiffness, yield stress and strength. App A will be used to optimize a set of mechanical properties of importance for the SME and will predict the target structure. - App B: a computational tool that relates mechanical properties of nonwovens to the type and pattern of (thermal or needlepunching) bonding while accounting for the network structure. This tool will be used by the SME to define the bonding pattern for target properties of nonwovens. - App C: an ML tool that provides information about the distribution of liquids in the network accounting for the network structure, fiber and liquid physical properties. This tool will be used by the SME to advance their application. RPI will interact with the SME to prioritize the tasks to best address the application(s). The industrial impact and commercialization are related to the application and remain the responsibility of the SME.

6. References

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization